

Literature Review Graphic Organizer
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Article	Author	Date	Website	Comments/Ideas	Page #	Quotes
The Relevance of Algorithms	Gillespie, Tarleton	2014	MIT Press	- Algorithms determine how “relevant” information is through rankings that are shaped by the priorities of institutions and the financial interests of their owners. - Systems of recommendation influence what people see when searching online, therefore creating what ultimately becomes “popular”. - When creators realize that algorithms are “filtering”, they will develop their material based on what they believe the algorithm will reward with exposure to the public. - The process described in the last paragraph also appears in the	167	“Algorithms play an increasingly important role in selecting what information is considered most relevant to us, a crucial feature of our participation in public life.” “Google’s algorithm recognized this—and mistakenly presumed it meant the search query “she invented” was merely a typographical error.” “Access to, understanding of, and rights regarding the algorithms that play a crucial role in public discourse and knowledge will likely change, for different stakeholders and under specific circumstances—changing also the power to build for, navigate through, and regulate these algorithms available to these stakeholders and those they represent.”

				work of Bishop and Bucher, as a behavioral adaptation of filtering. • Ranking can be thought of as an automated form of editorial power that determines what influences our perceptions of what is important. • Because it influences what we are exposed to collectively, ranking influences what we perceive as important, and subsequently affects what we participate in culturally. • As stated previously, the ranking process quietly changes what is created, and this source offers us the terms to describe the process.		
Want to be on the top? Algorithmic power and the threat of invisibility on Facebook	Bucher, Taina	2012	New Media & Society	• Bucher mentions a “risk of being invisible” for creators who are concerned with losing access to exposure at any time by the	6	• “The algorithm is not merely modelled on a set of pre-existing cultural assumptions, but also on anticipated or futureoriented assumptions about valuable and profitable interactions that are ultimately geared

				platform with little to no explanation; • This ambient concern is the cause for monitoring/adjusting behavior as a result of this concern. • Algorithmic power operates through anticipation and people act upon their beliefs about what the algorithm will reward them. • Creators regularly review their analytics data, and make adjustments based on assumptions that may or may not actually reflect the actual workings of the algorithm. • Creators react to perceived working of the algorithm, and these perceptions create self-sustaining systems of beliefs. • Creator communities have been shown to struggle with stability regarding visibility; however, this issue has not received enough		towards commercial and monetary purposes.” • “The media represent key mechanisms in the sorting, classification and ranking of the social field.”
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				attention from researchers focused on the technical aspects of platforms.		
“I always assumed that I wasn’t really that close to [her]”: Reasoning about invisible algorithms in the news feed	Eslami et al.	2015	ACM CHI Proceedings	• More than sixty percent of those in his study did not understand that algorithms were being used to filter what appeared in their news feeds on Facebook. • Missing content led people to make social interpretations (e.g., “we are no longer friends”) rather than realizing it may have been filtered out by the algorithm. • The users’ perceptions are being altered by the algorithm regardless of whether the user has knowledge of the algorithm itself. • The misunderstanding creates practical consequences as users construct theories about their social world	2	“Research attention has recently turned to algorithms as invisible and influential aspects of daily digital life. Many researchers have considered curation algorithms and argued that their effects are important while their operation is opaque [2, 19, 36].”

				using curation they believe to be complete. • It makes me wonder to what extent this phenomenon exists with regard to YouTube creators misinterpreting their own performance metrics. • This provides empirical evidence for the argument that transparency in these types of systems can lead to real-world implications. • There appears to be a vast difference between what is occurring technically in these systems, and what people perceive to be occurring.		
Google Search: Hyper-visibility as a Means of Rendering Black Women and Girls Invisible	Noble, Safiya Umoja	2013	InVisible Culture	• Noble says that search ranking algorithms create social hierarchies when they give priority to some ways to represent people over other ways to represent them.	3	• “Traditional media misrepresentations have been instantiated in digital platforms like search engines, and web search itself has been interwoven into the fabric of American culture.”

				• Noble illustrates this using a particular example involving search results that are potentially harmful to Black girls and women. • Ranking has implications in terms of both culture and politics. • Not everyone experiences visibility the same way; thus, while search ranking algorithms may contribute to greater equality in some cases, they can also exacerbate pre-existing inequities. • That hyper-visibility can function as invisibility is another idea I think will stick with me. • I am still trying to determine a thoughtful way to include identity elements into my analysis. • I believe the central theme of how ranking		
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				creates cultural perception is an important one.		
Managing visibility on YouTube through algorithmic gossip	Bishop, Sophie	2019	New Media & Society	• Bishop refers to “algorithmic gossip” as the informal theories or strategies that users develop regarding how recommendation systems function. • Since there isn’t much information available on how these systems actually operate, users develop common knowledge through group discussion in their online communities, which also limits access to official documentation. • Once visibility management becomes a normal social activity within communities, it’s not uncommon for those theories to be developed based on anecdotal evidence and very little data, resulting in less	Abstract	“Gossip is productive: community communication and talk informs and supports practices such as uploading frequently and producing feminised beauty content to perform more effectively on YouTube.”

				than accurate theories; • However, when the theories begin to take hold in a community and create a behavioral response, they become part of the user's interpretative culture regarding a system the user cannot completely understand. • In my view, opacity creates a knowledge ecosystem for users surrounding recommendation systems that I believe is one of the most fascinating aspects of this research.		
Deep Neural Networks for YouTube Recommendations	Covington, Paul; Adams, Jay; Sargin, Emre	2016	ACM RecSys	• This paper utilizes a two-phase video ranking system. The candidate generation phase limits a very large number of videos to a smaller set, and in the second stage, a ranking model assigns each video a score based on its	7	"We have described our deep neural network architecture for recommending YouTube videos, split into two distinct problems: candidate generation and ranking."

				expected watch time (i.e., how long users will view the video). • The output of the ranking model used to determine user exposure is produced from machine learning algorithms trained on historical behavioral data. • Watch time is the primary optimization target of the ranking model, and this produces an influence cascade effect in that as the ranking model predicts watch time based on past performance metrics (retention/duration), creators are incentivized to create content with high levels of retention and duration. • As the signals for retention and duration are also used as inputs into ranking model predictions, there can be a compounding		
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				effect of prior exposure influencing future exposure. • The process or logic that engineers use to make decisions about the presentation of video content to users is typically opaque, and this paper provides a mechanism for understanding this decision making process (i.e., makes it legible). • I believe that examining the relationship between the system design (architecture) of YouTube and how this affects the behavior of content creators may be one of the most significant ways to explore the implications of the paper's findings.		
The Socio-technical Architecture of Digital Labor	Postigo, Hector	2016	New Media & Society	• Postigo describes how platform developers take leisure-based content (i.e., gaming, creating	Abstract	"The main analytical lens is that of "affordances." It is used to map how technological features designed into YouTube create a set of probable uses/meanings/practices for users while serving YouTube's business

				videos) and convert it to resemble a form of labor. • Incentivizing users through various types of algorithms (e.g., recommender systems), as well as financial means (monetization infrastructure), allows platforms to leverage user desire for exposure (visibility) with an economic structure based on revenue. • Users are also guided towards desired behaviors through platform-based analytics, and through their monetization eligibility rules. However, these methods may not be immediately visible to the user. • Over time, these metrics influence the creative decisions of the user. • As I continue to think about this, I am trying to		interests.”
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				determine how much of this pressure is inherent to the platform's design versus how much of this is simply how incentivized systems function. • Socio-technical architecture is a useful term for describing the connection between how a platform is designed and how creators behave. • Creating content can lead to the embedding of labor expectations into the design of a platform because of the eligibility requirements and the location of recommendations within the system.		
The algorithmic dance: YouTube's Adpocalypse and the gatekeeping of cultural content on digital platforms	Kumar, Sangeet	2019	Internet Policy Review	• Kumar discusses the Adpocalypse, a situation that occurred in 2017 with an advertising backlash against YouTube due to brand safety issues and an eventual loss of ad	Abstract	"The March 2017 advertiser revolt on YouTube, popularly known as the adpocalypse, introduced widespread and radical changes on the platform's policies related to the moderation of content, their 'monetisability' and the terms of the relationship between creators and the platform."

				spend. • This created changes to monetization rules, along with categories of content as being “advertiser friendly” (or not). • Adpocalypse was a case of external market pressures applied to YouTube’s platform by its advertisers which changed the overall visibility of creators. • YouTube responded to changing commercial relationships by making policy changes for creators; and, creators were forced to make adjustments to their content based on these new policies. • The term “algorithmic dance” represents the cyclical process that exists with policy and rule changes made to the algorithms which affect creator visibility		
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				(and vice versa) until the next round of changes occur. • Visibility on platforms such as YouTube are influenced by the way the platform responds to commercial interests through policy changes that can be rapidly made. • Many times these changes appear to be arbitrary to the people they affect. • I am still trying to determine what relationship exists between the advertising backlash and the design of algorithms within the argument I am developing.		
Methods and metrics for cold-start recommendations	Schein, Andrew I.; Popescul, Alexandrin; Ungar, Lyle H.; Pennock, David M.	2002	ACM SIGIR	• The cold-start problem is a problem of the ability to recommend new content when there is insufficient interaction data. • Collaborative filtering relies upon prior interaction data	7	“However, in situations where the evaluation goal is to characterize the performance of a recommender system generally, both GROC and CROC curves are needed since they are often conflicting measures of recommender system performance.”

				for prediction. • A new video or creator entering the platform provides virtually all of its information for the model to operate from; this creates a structural disadvantage. • The structural disadvantage is independent of content quality. • This is a technical reason that explains why many new creators experience low visibility immediately after launching their product. • Low visibility can be due to the limited amount of data available to the algorithm; this is a technological limitation with practical consequences. • It also helps explain why creators focus so much on early metrics (such as number of views) and will attempt to generate traffic immediately after		
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				a video upload; the early traffic provides a signal for the algorithm. • I have been pondering whether the cold-start problem should be viewed solely as a technical limitation or as a fairness issue, and perhaps these issues are one and the same.		
Feedback Loop and Bias Amplification in Recommender Systems	Mansoury, Masoud; Abdollahpour, Himan; Pechenizkiy, Mykola; Mobasher, Bamshad; Burke, Robin	2020	arXiv	• Recommendation systems build a loop of feedback based upon what users are engaging with (popular items) to recommend them again, and generate more engagement data, which further reinforces their ranking. • Only a few products have all of the exposure. • The inequity of visibility compounds over time as the system learns from each of the outputs generated by itself. • Some of the data that the model is	Abstract	“Recommendation algorithms are known to suffer from popularity bias; a few popular items are recommended frequently while the majority of other items are ignored. These recommendations are then consumed by the users, their reaction will be logged and added to the system: what is generally known as a feedback loop.”

				being trained on was created by the model itself, and while it may be unsettling to ponder such an idea; it also seems to be creating some type of feedback loop in relation to the outputs that it generates. • The model has likely amplified or reinforced many of its own initial decisions, and in doing so, may appear to users to be organic preferences of the users themselves. • I believe the name we give to this cycle is important to how we articulate the structural aspect of inequity in visibility. • This self-reinforcing cycle provides barriers to upward mobility for emerging creators in tangible forms..		
Fairness of Exposure in Rankings	Singh, Ashudeep; Joachims,	2018	arXiv	• Singh and Joachims see exposure as a	4	“Virtually all utility measures used for ranking evaluation derive the utility of the ranking from the

	Thorsten			quantifiable resource that ranking systems provide and they develop a theoretical model to evaluate the fairness of this allocation. • Ranking systems can impact both users (people) and content (items), and therefore it is reasonable to consider how resources such as attention are distributed. • Therefore, if we consider whether ranking systems operate fairly, then we will add an evaluation component to the analysis. • Additionally, this could allow us to create systems where fairness requirements are considered simultaneously with user engagement maximization. • Honestly I do not know if that would be possible to implement on a large scale.		relevance of the individual items being ranked. “
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				The idea of viewing visibility as something that can be audited conceptually seems important.		
Algorithmically recognizable: Santorum's Google problem, and Google's Santorum problem	Gillespie, Tarleton	2017	Information, Communication & Society	- Gillespie identifies the ways users attempt to become legible to search and ranking algorithms by creating their content so as to mirror the algorithms' evaluation criteria. - Visibility involves learning to recognize what the algorithm recognizes and producing content based on that recognition. - Recognition by an algorithm is often a requirement for participating in these environments. - Quality is irrelevant if a piece of content does not meet the patterns that the system was trained to find; regardless of the content's quality it will not be visible.	6	"There is a distinction often made, between coordinated efforts to 'game' a search engine (like Googlebombing and SEO tactics) and the 'genuine' output of independent web producers and users, demonstrating the value of a site from their linking and clicking behaviors. The distinction is a false one. Most contributions to the web are somewhere in the middle, where people in some way coordinate their efforts in order to help make their content visible to a search engine, out of a 'genuine' desire for it to be seen. Activist organizations publicize their efforts and gather supporters online; companies urge their customers to post about their product, sometimes in exchange for rewards; writers send their posts to their friends or tweet them to their followers"

				• This creates pressure for creators to produce content in machine-readable formats. • Users can't always articulate the pressure they feel to create in machine-readable formats due to platform infrastructure that defines what behavior is rewarded with visibility.		
The New Governors: The People, Rules, and Processes Governing Online Speech	Klonick, Kate	2018	Harvard Law Review	• Klonick argues that large platforms function similarly to governance bodies. They do this through moderation of speech through their moderation policies, as well as content curation of speech they deem appropriate. • Private companies are taking on the roles of public institutions. • The recommendation and ranking systems of platforms operate		“Facebook blamed the takedown on a “technical glitch” but provided no further details. This is not entirely surprising. Though it might appear that any internet user can publish freely and instantly online, many content-publication platforms actively moderate the content posted by their users. Yet despite the essential nature of these platforms to modern free speech and democratic culture, very little is known about how or why these companies curate user content.”

				in governance type structures which ultimately decide who will have their voice heard most clearly. • Klonick examines policy enforcement, content removal, and internal decision-making processes of the platform itself. • The authority of the platform has direct and measurable effects on the level of discourse among participants. • The concept that platforms exert governance-type influence over discourse adds a normative aspect to the visibility issue. • I am unsure of how much I should extend the comparison, And I want to be cautious of extending too far the analogy.		
Free Speech in the Algorithmic Society	Balkin, Jack M.	2018	UC Davis Law Review	• The term “Algorithmic Society” is used by	8	“In the Algorithmic Society, surveillance and data collection are now

				Balkin to describe the manner in which communication is structured using automated systems and collecting large amounts of data. • The private companies that build and control social media platforms have tremendous influence over the flow of information via their own architecture (infrastructure) and by controlling what we see (ranking). • Balkin suggests that recommendation algorithms can also impact democratic processes as they determine what we see/hear when engaging with platforms. • While he expands his argument beyond just the technical into legal/constitutional areas, • I am likely to stay away from the		widely distributed, but there is no guarantee that they will be democratically controlled. Data about many people are collected in many places, but a relatively small number of people have the resources and the practical ability to collect, analyze, and use this data.”
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				legal aspects of this issue in my own paper. • However, having an understanding of the legal issues allows me to connect concerns around visibility to governance and accountability in ways that are relevant and interesting to me. • Lastly, there are very real implications for power structures surrounding how these systems function and I would like to at least mention them.		
What Do We Mean When We Talk About Transparency? Toward Meaningful Transparency in Commercial Content Moderation	Suzor, Nicolas P.; West, Sarah M.; Quodling, Amanda; York, Jillian	2019	International Journal of Communication	• Suzor et al. state that in order for transparency to exist, there must be clearly defined rules/decision processes, as well as an appeals process. • In order to have meaningful accountability, a platform must provide a substantive explanation of its decision-making	2	“In this article, we directly address the conceptual problem of vagueness in calls for transparency in the moderation of social media content. We argue that there is a pressing need for more specificity in identifying what information should be provided and to whom.”

				processes. • Transparency is hindered by the lack of information regarding the ranking systems on platforms which makes it difficult for users to understand the reasons behind their visibility results. • This relates to the Algorithm Misunderstanding thread as it highlights several specific areas where the user is misinformed or confused and leads to speculative thinking. • The authors are able to provide a description of what actual transparent practice would look like in regards to ranking; which provides a set of normative criteria that can be used to evaluate the practices of each platform. • A better communication strategy for		
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				explaining the logic of ranking systems may alter how creators view and react to the system. • Although I do not believe that the platforms will completely embrace the type of transparency described, having a normative standard to refer to will likely be beneficial in solidifying the argument.		
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